



CHEMISTRY



ORGANIC



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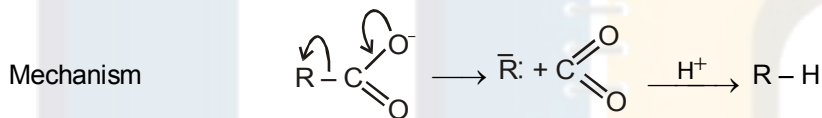
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1. Alkane

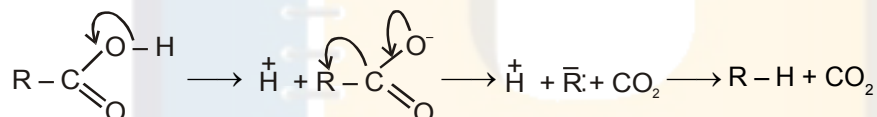
Wurtz reaction :



Decarboxylation :

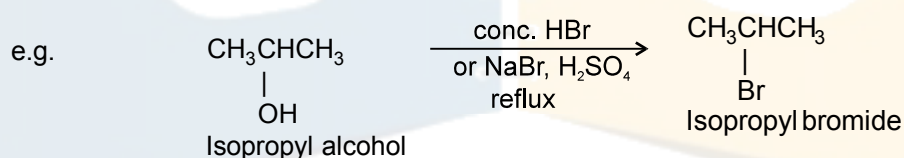
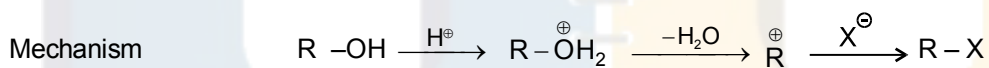
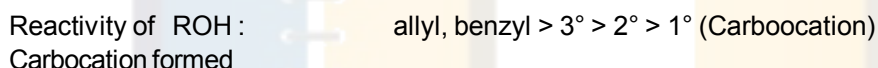
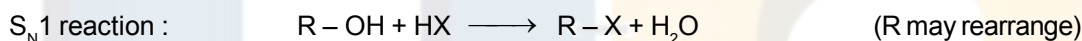


The thermal decarboxylation of free acids may be :

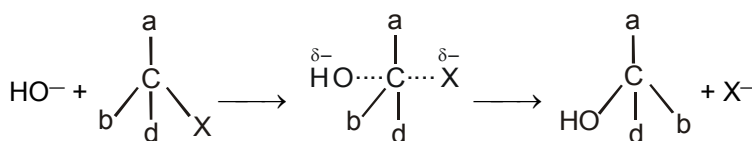
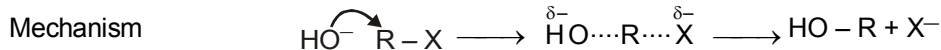
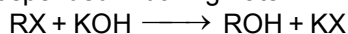


2. Alkyl halide

Nucleophilic substitution Reaction (S_N1 , S_N2)



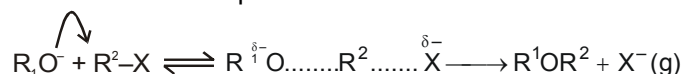
S_N2 reaction : Alkylhalide are hydrolysed to alcohol very slowly by water, but rapidly by silver oxide suspended in boiling water.





Williamson's synthesis :

It is the reaction in which sodium or potassium alkoxide is heated with an alkyl halide (S_N2).



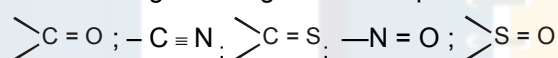
This method is particularly useful for preparing mixed ethers.

3. Grignard Reagents

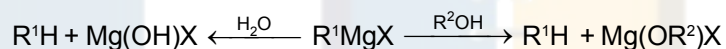


The majority of Grignard reactions fall into two groups

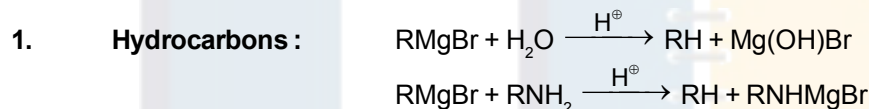
(i) Addition of the Grignard reagent to a compound containing a multiple - bond group e.g.



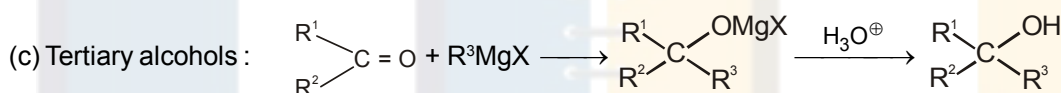
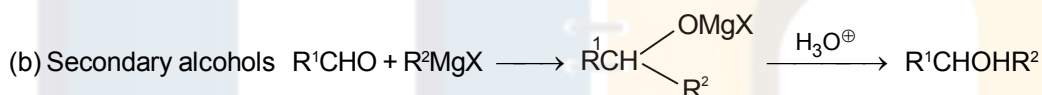
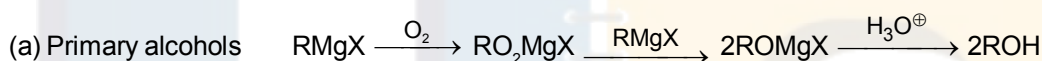
(ii) Double decomposition with compounds containing an active hydrogen atom or a reactive halogen atom.



Important chemical synthesis by Grignard reagent



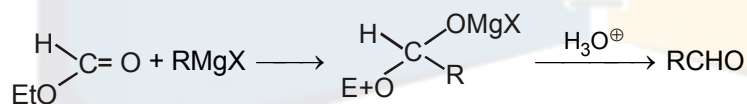
2. **Alcohols :**



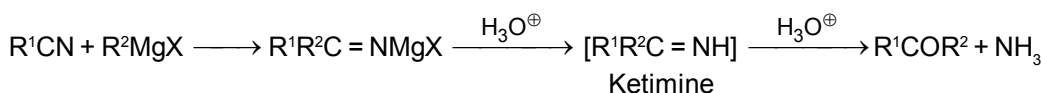
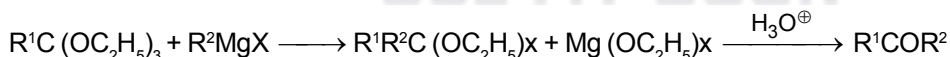
3. **Ethers :**



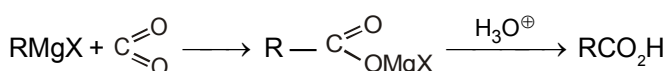
4. **Aldehydes :**



5. **Ketones :**

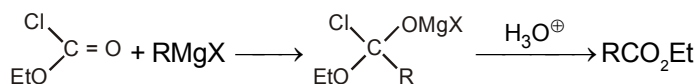


6. **Acids :**

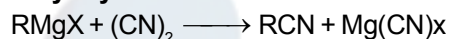




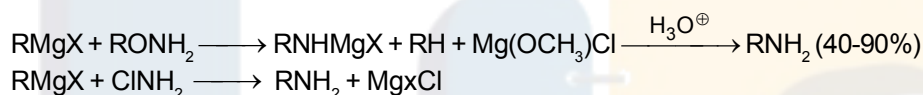
7. **Esters :**



8. **Alkyl Cyanides :**



9. **Primary amines :**



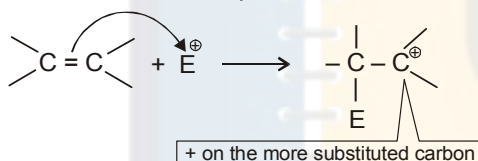
4. Alkene & Alkyne

Electrophilic addition reactions :

Mechanism

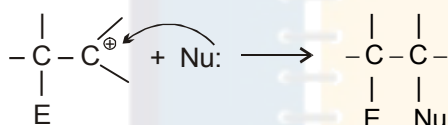
Step 1 :

Attack of the electrophile on π bond forms a carbocation.

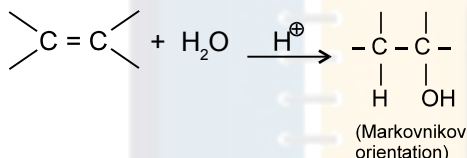


Step 2 :

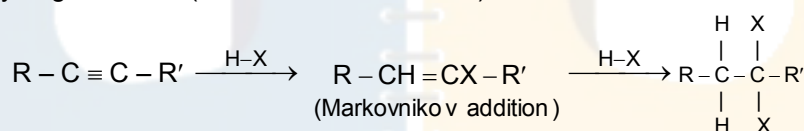
Attack by a nucleophile gives the product of addition.



e.g. (a) Addition of water



(b) Addition of hydrogen halides (where $\text{HX} = \text{HCl}, \text{HBr}, \text{HI}$)

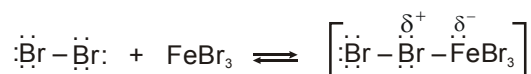


5. Aromatic compounds

Electrophilic aromatic substitution :

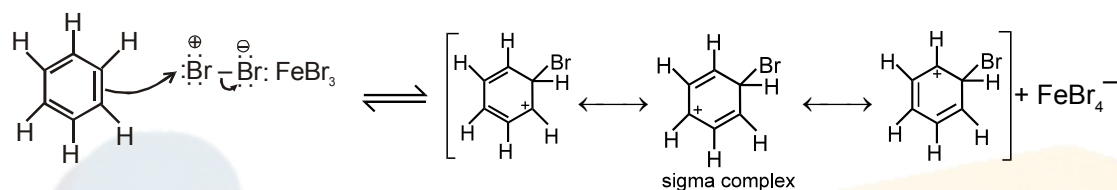
(a) Bromination of Benzene : Bromination follows the general mechanism for electrophilic aromatic substitution. Bromine itself is not sufficiently electrophilic to react with benzene, but a strong Lewis acid such as FeBr_3 catalyzes the reaction.

Step 1 : Formation of a stronger electrophile.

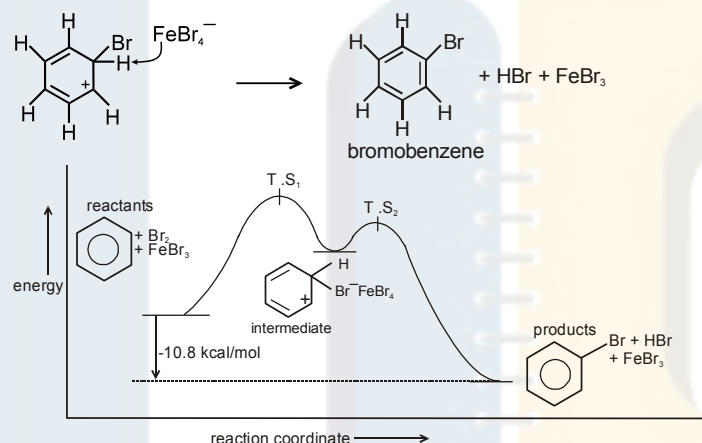


Step 2 : Electrophilic attack and formation of the sigma complex.

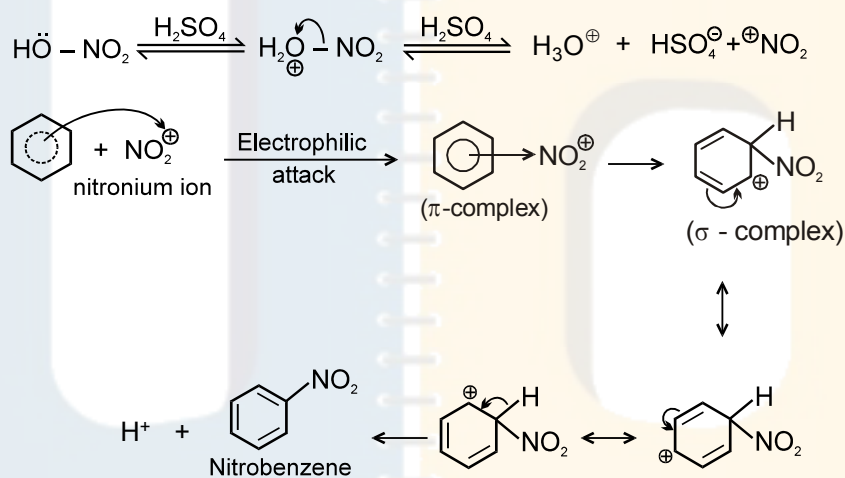




Step :2 Loss of a proton gives the products.

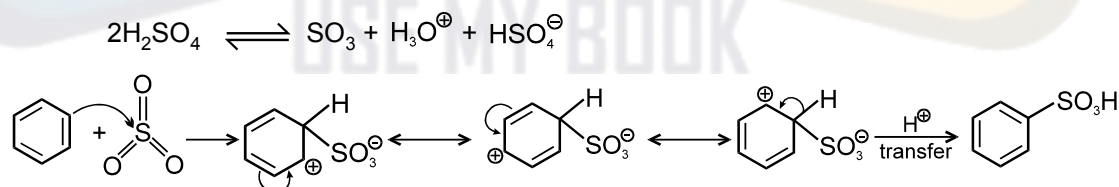


(b) Nitration



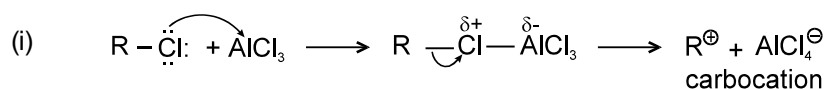
(c) Sulphonation :

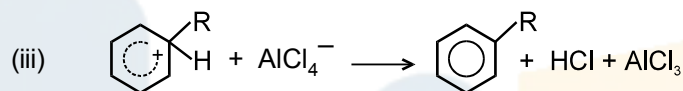
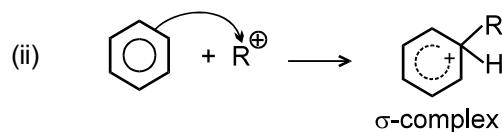
The electrophilic reagent, SO_3 , attacks the benzene ring to form the intermediate carbocation.



(d) Friedel Craft reaction :

Alkylation mechanism

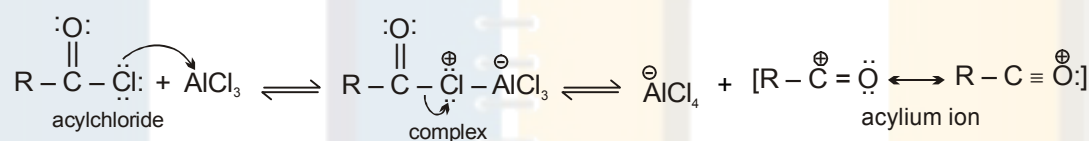




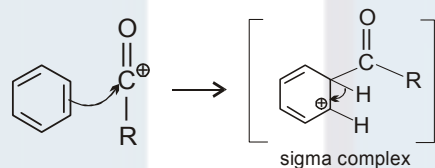
Acylation mechanism

Acylation of benzene may be brought about with acid chlorides or anhydrides in presence of Lewis acids.

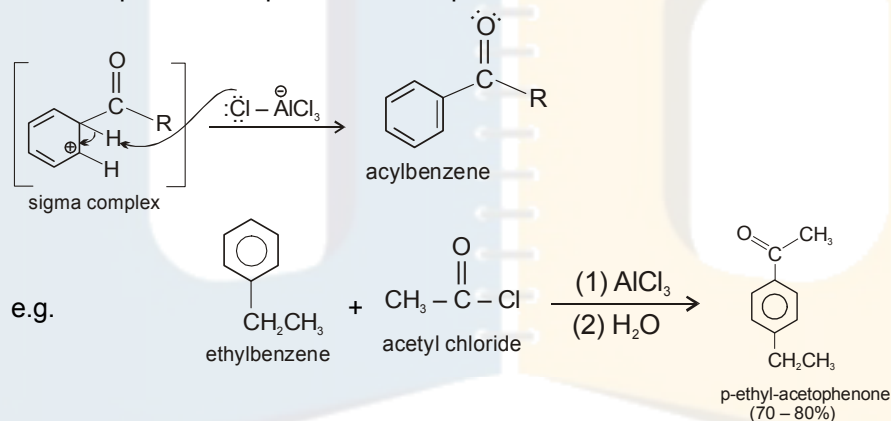
Step 1 : Formation of an acylium ion.



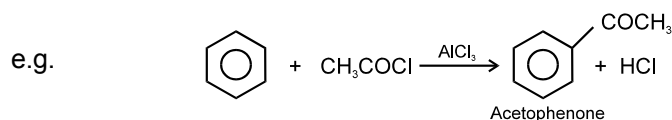
Step 2 : electrophilic attack.



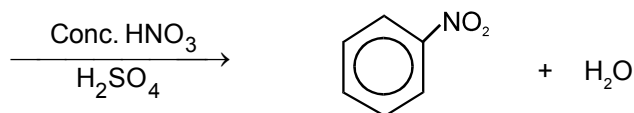
Step 3 : Loss of a proton. Complexation of the product.

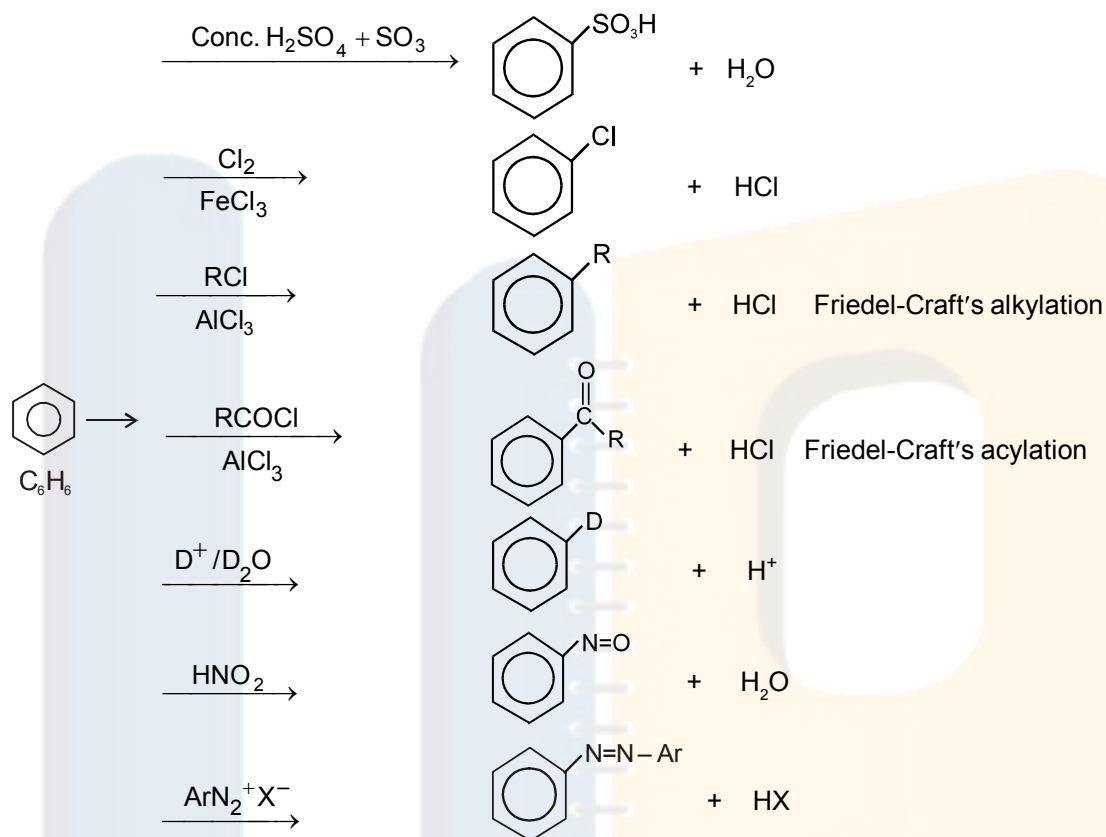


Note : Friedal - Crafts acylations are generally free from rearrangements and multiple substitution. They do not go on strongly deactivated rings.



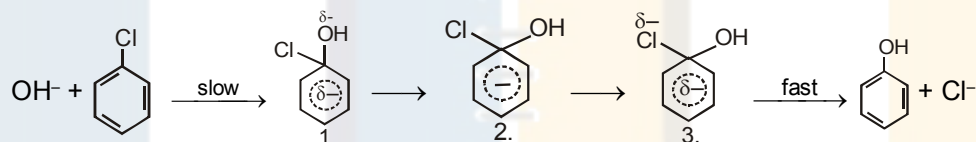
Chemical Reactions of Benzene :





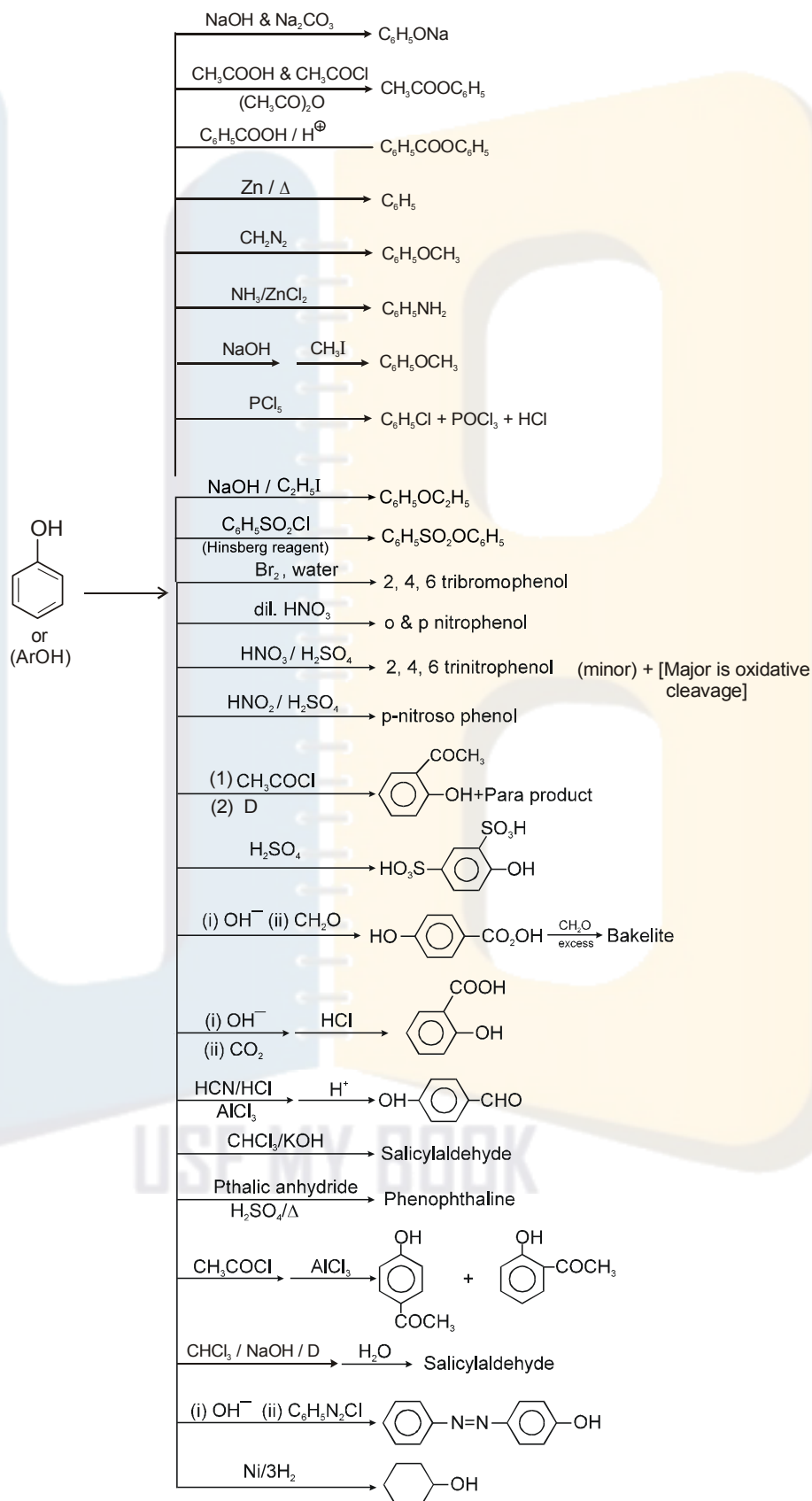
Nucleophilic Aromatic Substitution :

The reaction is second-order in which nucleophilic substitution occurs on benzene ring. It is generally accepted that the reaction proceeds via an intermediate σ -complex, the benzenonium carbanion (or the pentadienyl anion), e.g.,





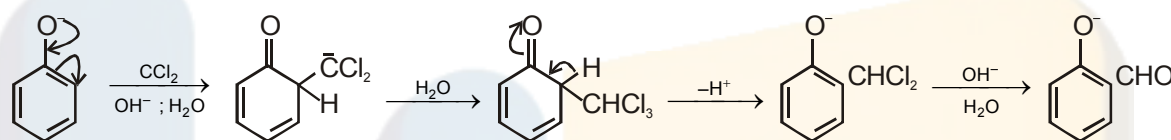
6. Chemical Reactions of Phenol





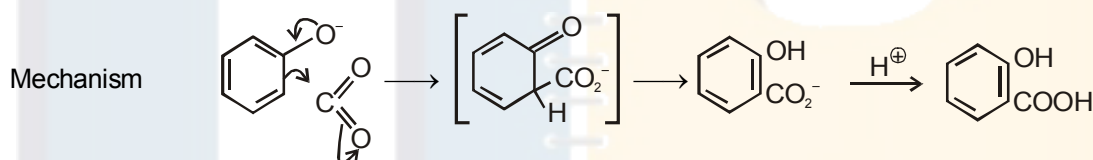
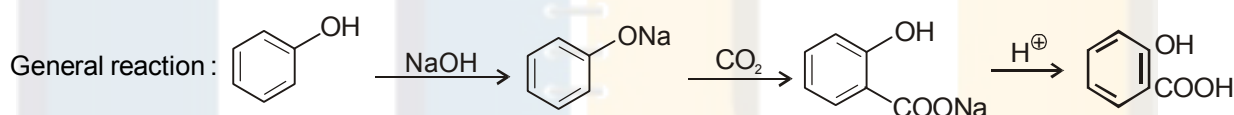
Reimer-Tiemann reaction :

The reaction is conversion of phenol to salicylaldehyde. The mechanism of the Reimer-Tiemann reaction is believed to involve the formation of dichloromethylene.



Kolbe Reaction :

It is the industrial method of preparation of salicylic acid from phenol.

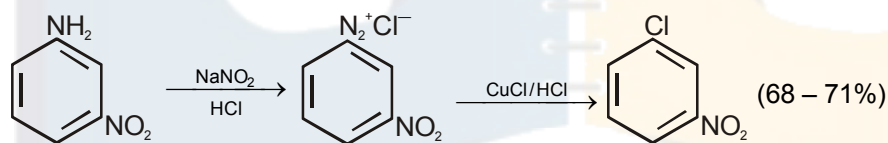


7. Chemical reactions of aniline

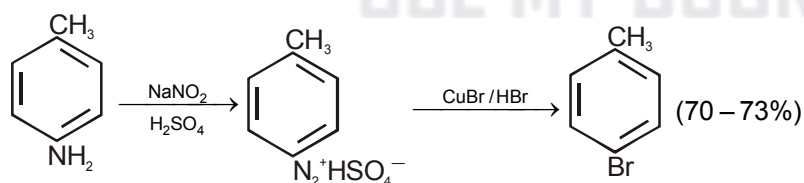
Sandmyer's Reaction :

When a diazonium salt solution is run into a solution of cuprous halide dissolved in the corresponding halogen acid, the diazo-group is replaced by a halogen atom.

m – Chloronitrobenzene

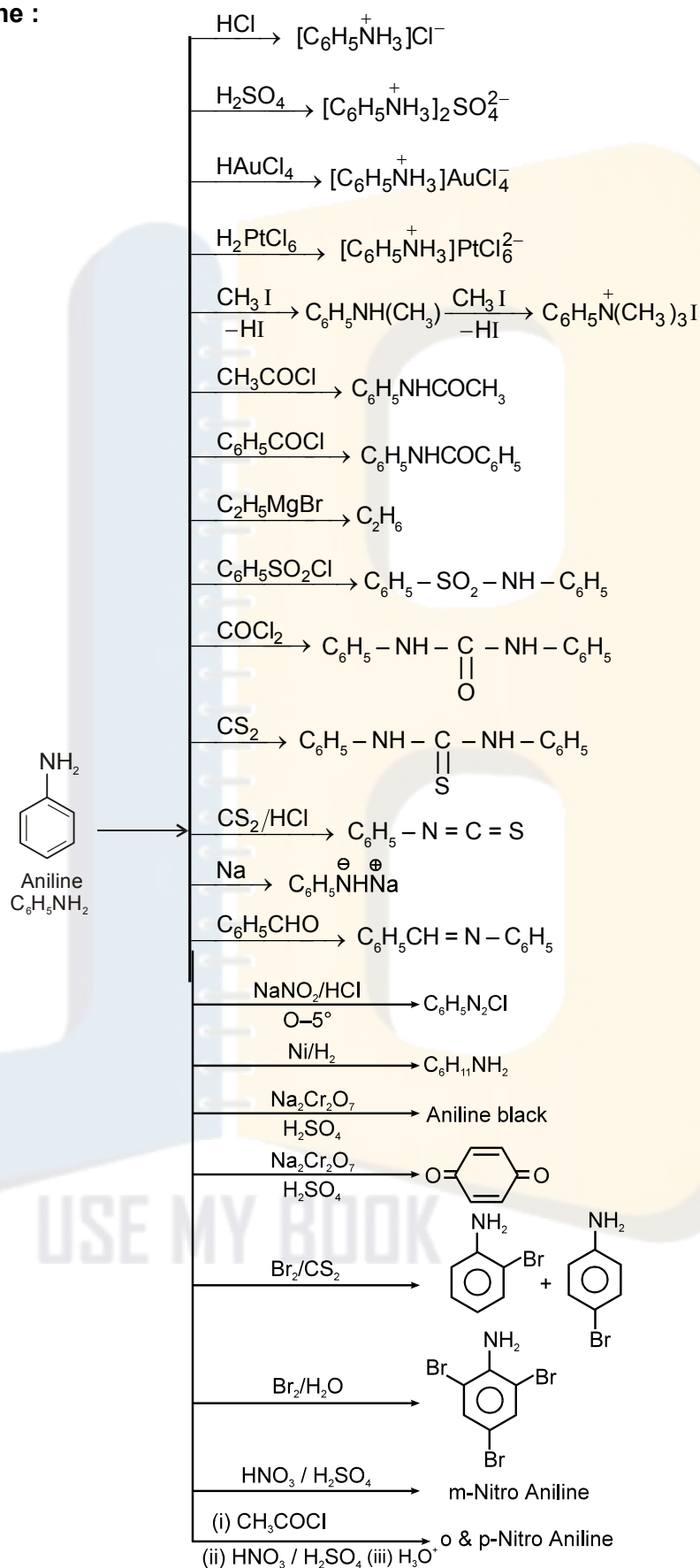


p – Bromotoluene





Chemical Reactions of Aniline :

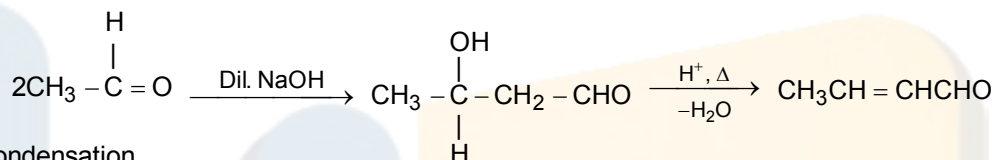




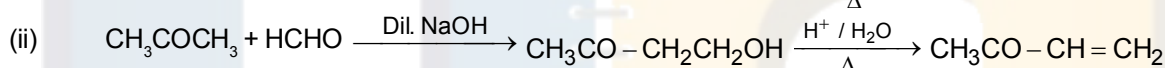
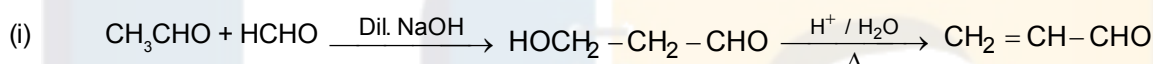
8. Aldehyde & ketone

Aldol condensation :

Carbonyl compounds having acidic α - H shows this reaction in presence of dil. NaOH or dil. acid.

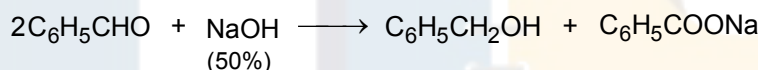


Crossed aldol condensation

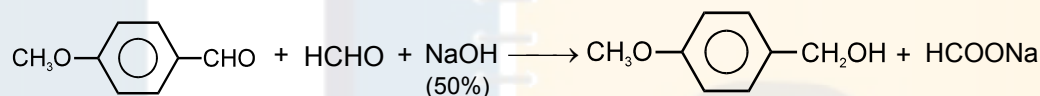


Cannizzaro reaction :

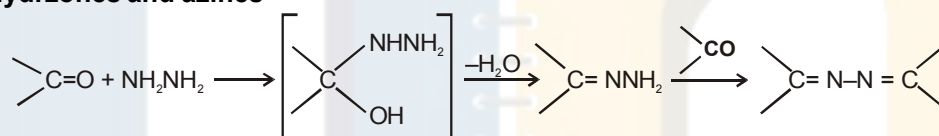
Carbonyl compounds not having α -H shows following disproportion reaction



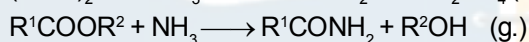
Crossed Cannizzaro reaction :



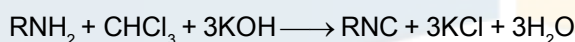
Formation of hydrazones and azines



Amides formation :

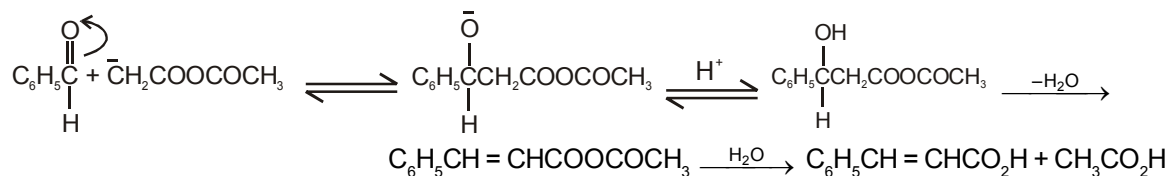
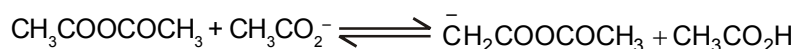
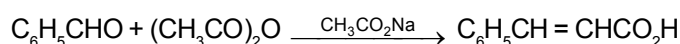


Carbyl amine reaction



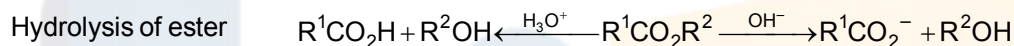
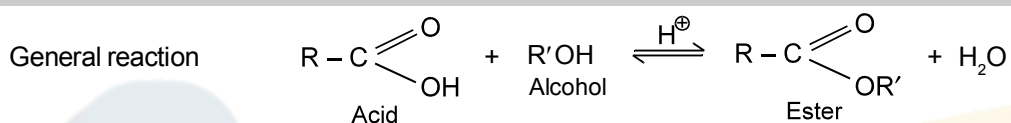
Perkin reaction :

When benzaldehyde (or any other aromatic aldehyde) is heated with the anhydride of an aliphatic acid (containing two α -hydrogen atoms) in the presence of its sodium salt, condensation takes place to form a β -arylacrylic acid ; e.g., with acetic anhydride and sodium acetate, cinnamic acid is formed.



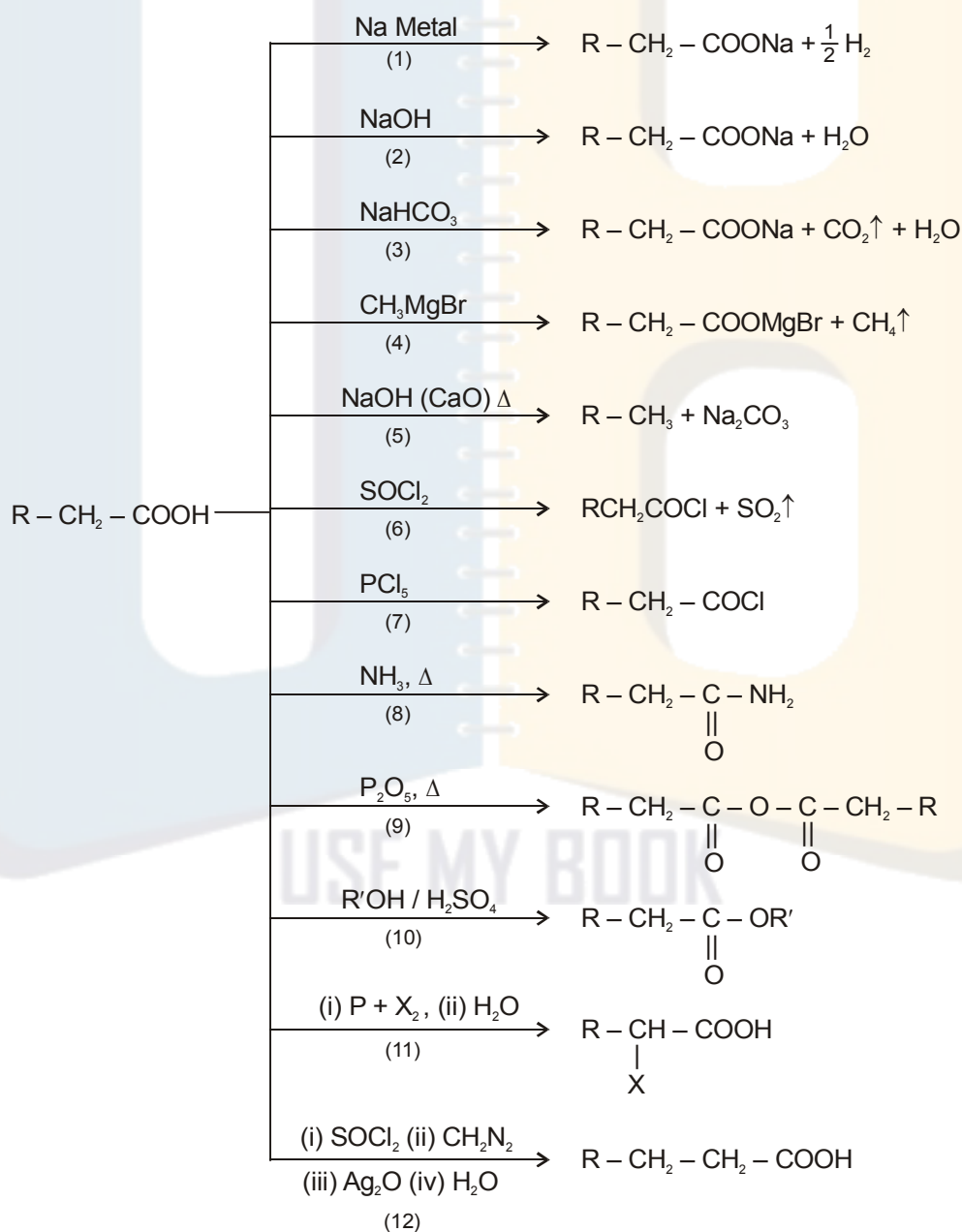


9. Ester formation



10. Carboxylic Acids

Chemical Reactions of acids





Identification of Functional Groups by Laboratory Tests (POC)

Table :

Functional Groups	Reagent	Observation	Reaction	Remarks
$C \equiv C$ / $C \equiv C$	[Bayer's reagent] alk. dil. cold $KMnO_4$	Pink colour Disappears	$CH_2 = CH_2 + H_2O + O \xrightarrow{\text{alk. } KMnO_4} \begin{matrix} CH_2 - CH_2 \\ \quad \\ OH \quad OH \end{matrix}$	Hydroxylation
	Br_2 / H_2O	Red colour decolourises	$Br_2 + CH_2 = CH_2 \longrightarrow \text{white ppt}$	Bromination
$R - C \equiv CH$	(a) Cuprous chloride + NH_4OH	Red ppt.	$R - C \equiv CH + CuCl \xrightarrow{NH_4OH} R - C \equiv C Cu \downarrow (\text{red})$	
	(b) $AgNO_3 + NH_4OH$	White ppt.	$R - C \equiv CH + Ag^+ \longrightarrow R - C \equiv C Ag \downarrow (\text{white})$	
ROH 3° 2° 1°	Lucas Reagent [Conc. HCl + anhyd. $ZnCl_2$]	(3°) Cloudiness appears immediately (2°) Cloudiness appears within 5 min. (1°) Cloudiness appear after 30 min.	$R - OH + HCl \xrightarrow{\text{anhydrous } ZnCl_2} R - Cl + H_2O$ cloudiness	Lucas Test I. ter. alcohol II. sec. alcohol III. pri. alcohol





Functional Groups	Reagent	Observation	Reaction	Remarks
Ar - OH Enols	FeCl ₃ (Neutral)	Coloured ppt. (violet, blue, green buff)	$6 \text{C}_6\text{H}_5\text{OH} + \text{FeCl}_3 \longrightarrow [\text{Fe}(\text{PhO})_6]^{-3}$	Test of enols / phenols
$>\text{C}=\text{O}$	2,4-Dinitrophenyl hydrazine (2,4-DNP) solution	Yellow orange ppt.	$>\text{C}=\text{O} + \text{H}_2\text{N}\cdot\text{NH}\cdot\text{C}_6\text{H}_3(\text{NO}_2)_2 \longrightarrow \text{C}_6\text{H}_3(\text{NO}_2)_2\text{NHNH}\cdot\text{C}(\text{NO}_2)_2 + \text{H}_2\text{O}$ $>\text{C}=\text{O} + \text{H}_2\text{N}\cdot\text{NH}\cdot\text{C}_6\text{H}_3(\text{NO}_2)_2 \longrightarrow \text{C}_6\text{H}_3(\text{NO}_2)_2\text{NHNH}\cdot\text{C}(\text{NO}_2)_2 + \text{H}_2\text{O}$ $\text{RCHO} + \text{Cu}^{+2} \xrightarrow{\text{Fehling's sol}^n} \text{RCOOH} + \text{Cu}_2\text{O} \downarrow + 2\text{H}_2\text{O}$ $\text{RCHO} + \text{Ag}^+ \xrightarrow{\text{RCCOH}} \text{RCCOH} + 2\text{Ag} \downarrow \text{ (Silver mirror)}$	DNP-test
R - CHO	Fehling solution A & B	Red ppt.	$\text{RCHO} + \text{Cu}^{+2} \xrightarrow{\text{Fehling's sol}^n} \text{RCOOH} + \text{Cu}_2\text{O} \downarrow + 2\text{H}_2\text{O}$	Fehling's test
	Tollen's reagent	Black ppt. or silver mirror	$\text{RCHO} + \text{Ag}^+ \xrightarrow{\text{RCCOH}} \text{RCCOH} + 2\text{Ag} \downarrow \text{ (Silver mirror)}$	Tollen's test
	Schiff's Reagent *	Pink colour resume		
R - COCH ₃ or ArCOCH ₃ or CH ₃ CHO	I ₂ / NaOH	Yellow ppt of CHI ₃ (iodoform)	$\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{I}_2, \text{NaOH}} \text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{ONa} + \text{CHI}_3 \downarrow$ $\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{I}_2, \text{NaOH}} \text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{ONa} + \text{CHI}_3 \downarrow$	Iodoform reaction
$\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{OH}$	Blue litmus	Litmus change to red.		Litmus test.
	Conc. NaHCO ₃ solution	Effervescence evolve.	$\text{R}-\text{COOH} + \text{NaHCO}_3 \longrightarrow \text{R}-\text{COO}^-\text{Na}^+ + \text{H}_2\text{O} + \text{CO}_2 \uparrow$	Sodium bicarbonate test

★ Schiff's reagent : p-Rosiniline hydrochloride saturated with SO₂ so it is colourless. The pink colour is resumed by RCHO.

